



Per-pixel land cover accuracy prediction: A random forest-based method with limited reference sample data

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ARTICLE INFO

Keywords:

Land cover mapping
Support vector machine
Spatial accuracy
Accuracy assessment
Random forest

ABSTRACT

Given the importance of accuracy in land cover (LC) maps, several methods have been adopted to predict per-pixel land cover accuracy (PLCA) of classified remote sensing images. Such a PLCA map provides spatially-explicit accuracy information and is of paramount importance for both producers and end-users of LC maps to thoroughly understand the spatial distribution of accuracy. In this study, we proposed a simple yet powerful random forest (RF) based approach for PLCA mapping with limited reference sample data. The main assumption of the proposed approach is that the LC's misclassifications do not occur randomly, but rather exhibit some detectable characteristics which can be retrieved via the built model. With this approach, RF attempts to establish a nonlinear relationship between the accuracy and the same spectral bands used in LC classification. To confirm the proposed method as a consistent and practical approach for a variety of different settings, we evaluated it on five different classified remote sensing images derived from Landsat-8, Ikonos, and three Sentinel-2 images across different parts of Iran. In this manner, to validate the predictive capability of the RF-based method, we calculated the area under the receiver operating characteristic curve (AUROC) and several other statistical metrics, including sensitivity (SN), specificity (SP), positive predictive value (PPV), negative predictive value (NPV), and accuracy (ACC). Analysis of the average values of these metrics (AUROC = 0.88, SN = 95%, SP = 68%, PPV = 96%, NPV = 72%, and ACC = 95%) derived from the limited sample size datasets showed that the proposed model performs well in all case studies. The performance of the proposed model was further assessed through comparison against two benchmark methods, namely Gaussian kernel interpolation (GKI) and linear kernel interpolation (LKI). In conclusion, although our comprehensive evaluations revealed that RF, GKI, and LKI methods are promising approaches for PLCA mapping, RF outperformed both GKI and LKI in all of the experimental sites.

1. Introduction

Land cover (LC) maps are among the most important sources of information in different fields such as natural resource management, environmental monitoring, climate change, environmental risk assessment, and land-change studies (Gómez et al., 2016; Khatami et al., 2016). In recent years, due to the importance and considerable rise in interest towards LC maps, among both researchers and end-users, extensive studies have been devoted to investigate various aspects of the production and evaluation of LC maps. Among these issues of interest, quality and accuracy of the LC maps have been considered as important research issues (Foody, 2002; Gahegan and Ehlers, 2000; Olofsson et al., 2014; Smith et al., 2003; Steele et al., 1998; Stehman and

Foody, 2019; Zhang et al., 2018).

All spatial data, including LC maps, due to the simplification of the reality are inevitably exposed to different levels of errors and uncertainties (Devlillers et al., 2010; Foody, 2002; Smits et al., 1999). Probable errors are accumulated across different stages of LC map production, including data acquisition, processing, analysis, and classification (Congalton, 1991; Stehman and Foody, 2019). The effects of these errors are potentially propagated to the final outputs and derived decisions from LC maps. Therefore, quantification and provision of accuracy metrics along with LC maps are essential. The reason for this necessity is twofold: an LC map without any information regarding its accuracy is no more than an unconfirmed hypothesis, and, more importantly, without accuracy information end-users will not be able to

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<https://doi.org/10.1016/j.isprsjprs.2020.11.024>

Received 19 May 2020; Received in revised form 16 November 2020; Accepted 23 November 2020

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